

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc (MI) Examination April -May 2019

Course code & Name: MI 862 Quality Control and Assurance

Date: 29/4/2019

Day: Monday

Time: 1:30pm To 3:30 pm

Maximum Marks: 35

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Part A

Time: 1:30pm To 1:50 pm

Maximum Marks: 10

Q – I Choose the correct answer for the following questions.

10

- 1) The following is usually not carried out by a quality control personnel
(i) sampling (ii) testing (iii) sale and marketing (iv) all of the above
- 2) Total Quality Management emphasizes most on the satisfaction of the needs and expectations of the following
(i) community (ii) employees (iii) leadership (iv) FDA
- 3) Preparations containing live microorganisms may be made or containers filled in areas having the following facility
(i) devices for dry heat sterilization (ii) sterile air under positive pressure
(iii) fumigation gases (iv) all of the above
- 4) A intravenous preparations should be free of the following validated by test for endotoxins
(i) pyrogens (ii) microorganisms (iii) chemical contaminant (iv) viruses
- 5) An effective quarantine minimizes the chances of introduction of the following into an established colony of laboratory animals in animal house
(i) stability (ii) pathogen (iii) good hygiene (iv) all of the above

P.T.O.

- 6)** Bacterial inoculums may be stored under refrigeration (2-8° C) for up to the following after preparation
(i) 1 day (ii) 2 days (iii) 7 days (iv) 2 hours
- 7)** Rinse fluid A supplemented Polysorbate 80, adjust to pH 7.1 ± 0.2 is required for Sterility test of products containing the following
(i) antibiotics (ii) antimicrobial agent (iii) oil (iv) surfactant
- 8)** Experiments involving indigenous or exotic agents with potential aerosol transmission; may have serious or lethal consequences may be tested under the following facility
(i) BSL I (ii) BSL II (iii) BSL III (iv) BSL IV
- 9)** A radionuclide in a kit of radiopharmaceutical preparation may be diluted in the following solution
(i) non-radioactive (ii) isotopic (iii) non-isotopic (iv) radioactive
- 10)** Friability Test is generally carried out for the following pharmaceutical products
(i) ointment (ii) transdermal patches (iii) tablets (iv) cream

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Instructions:

1. Section I and II must be attempted in same ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

Part B **Time: 1:50pm To 3:30 pm** **Maximum Marks: 25**

SECTION –I

10

Q – II A Write a short note on any two of the following: (06)

- (i) TQM
- (ii) CPCSEA guidelines for animal experimentation involving hazardous agents
- (iii) Protocol to Conduct a clinical laboratory study

Q – II B Discuss briefly any two of the following: (04)

- (i) ISO series with reference to quality of a product
- (ii) Bioburden estimation of medical devices
- (iii) Components of kit for radiopharmaceutical preparation

SECTION – II

15

Q – III A Attempt any two of the following: (06)

- (i) List the key SOPs required in a testing facility. Discuss SOP for handling OOS results in brief
- (ii) Discuss briefly FPQC tests for pharmaceutical tablets
- (iii) What is the importance of sterility testing of pharmaceutical products? Discuss system control in brief

Q – III B Why should intravenous injectable be free of pyrogens? Discuss the LAL test in detail. (03)

OR

Q – III B Discuss briefly: Clean Room Activities during sterility testing of a pharmaceutical product. (03)

Q – IIIC Discuss briefly any two of the following: (06)

- (i) Bioassay of antibiotics in pharmaceutical product
- (ii) Suitability of counting method in Antimicrobial effectiveness test.
- (iii) The Standard Microbiological Practices for various Biosafety Levels in biomedical and microbiology Laboratory